Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/23/2006
PI Name:	Goins, Gregory D Ph.D.		
Project Title:	Crop Production Technology: Advanced Life Support		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	336-334-7907
Organization Name:	North Carolina A&T University		
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City:	Greensboro	**State:**	NC
Zip Code:	27411	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	12/12/2003	**End Date:**	01/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Raymond M Wheeler, PhD, NASA Biological Sciences Office, Kennedy Space Center, FL is replacement PI; Dr. Goins moved to NC A&T Univ. Research team: Nate Cranston, Sharon Edney, Jeff Richards, Gary Stutte, Neil Yorio, Dynamac, Kennedy Space Center		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The first opportunities for using plants in life support systems for space will likely involve a vegetable production unit (VPU) or “salad machine” to provide fresh vegetables and fruits to supplement the crew’s diet. Several crops were identified for initial testing, including: tomato, spinach, radish, chard, lettuce, cabbage, carrot, and green onion. To assess the challenges and potential yields in a typical space cabin environment, radish, lettuce and onion plants were grown at three temperatures (22, 25, and 28 C), three carbon dioxide concentrations (400, 1200, and 4000 ppm), and three light intensities (150, 300, and 450 micro mol m-2 s-1). A similar series of tests was carried out with dwarf tomato and dwarf pepper but only at 25 C. All the studies were repeated to provide statistical confidence, and samples were analyzed for taste and acceptability attributes by taste panels. Related studies were carried out with radish and lettuce just at 25 C, 1200 ppm carbon dioxide, and several light		

	intensities but in this case using light emitting diodes (LEDs). LEDs are solid state lamps that have a long operating life and hence are appealing for space applications. In addition, LEDs emit very little long wave radiation (heat) toward the plants, simplifying thermal management issues for space chambers. But the spectra of LEDs is very narrow compared to conventional fluorescent and high-intensity discharge lamps, and plant responses to these narrow spectra need study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Findings from NASA sponsored testing with salad crops will provide valuable information to commercial controlled environment agriculture (CEA) operations, where optimizing light, carbon dioxide and temperature are important for maximizing yields. The routine use of recirculating hydroponic systems in NASA studies will also provide useful information for reducing nutrient discharge from commercial CEA systems into ground water and nearby aquatic systems. LED development continues at a rapid pace in the commercial world (e.g., traffic lights and car tail lights), and LEDs have already been tested in commercial CEA facilities in Japan. NASA's continued testing with crops grown under LEDs will provide valuable information needed to develop acceptable combinations of spectra and intensity for competitive and profitable CEA systems around the world.
Task Progress:	Radish, lettuce, and green onion plants were grown hydroponically for 35 days at different combinations of temperature (22, 25, and 28 degrees C), carbon dioxide concentration (400, 1200, and 4000 ppm), and light intensity (150, 300, 450 umol m-2 s-1). Each test was repeated twice and harvest weights, texture, and color were assessed. A similar series of tests was conducted with dwarf tomato and dwarf pepper where plants were grown for 105 days at 400, 1200, and 4000 ppm carbon dioxide, 150, 300, and 450 umol m-2 s-1 light, but only at 25 degrees C temperature. For the tomato and pepper studies, fruits were also compared for sensory (taste) analysis. In general, increased light intensity increased growth for all species, but also increased the incidence of leaf tipburn injury in lettuce. Likewise, increasing the CO2 concentration to 1200 ppm from 400 ppm increased the growth for all species. Growth of lettuce and radish plants under red LEDs alone typically resulted in elongated stems and leaves, and occasional bleached (no chlorophyll) areas in radish leaves. Added blue light to the red prevented these effects and resulted in acceptable growth. Addition of green light to the red and blue increased growth in some cases but had no effect in others. But in all cases, the addition of green light provided a more balanced spectrum that allowed visual assessment of the health and development of the plants. When green light was provided alone, stomatal conductance of lettuce plants was reduced, suggesting a direct effect of the green light on the stomatal control mechanism.
Bibliography Type:	Description: (Last Updated: 03/20/2017)
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